Shreyan Balaji Nalwad

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EDUCATION

Carnegie Mellon University — Pittsburgh, USA
Master of Science in Computational Biology — Dec 2026

Vellore Institute of Technology — Vellore, India
Bachelor of Technology in Computer Science and Engineering, Bioinformatics concentration — May 2025

SKILLS

Computer Vision and ML: PyTorch, CUDA, segmentation and object detection (YOLO, Cellpose, StarDist), generative models (DDPM, VAE, U-Net), foundation models (UNI2, scGPT), ViT, adversarial training, active learning, GNNs.
Imaging and Spatial Omics: expansion and multiplexed fluorescence microscopy (ND2, HDF5), WSI (OpenSlide, Napari), Visium (scanpy, AnnData), MALDI-MSI (pyimzML, METASPACE), scRNA-seq (Seurat).
Systems and Deployment: Edge computing, Weights & Biases, NVFlare, AWS (Certified Cloud Practitioner), S3, Git.
Languages: Python, R, Go, C++, Java, SQL.

EXPERIENCE

Zhao Biophotonics Lab, Carnegie Mellon University — Pittsburgh, USA
Graduate Research Assistant — Jan 2026 – Present

- Developed **NuStain**, an **in silico multiplexing** method predicting antibody stains from a single NHS ester total protein channel of expansion microscopy tissue, across a **10 marker renal panel** in human kidney and **12 neuronal antibodies** in mouse brain, to cut reagent and imaging cost. *First author manuscript in preparation.*
- Built a **leakage free evaluation framework** (held out tissue cases, guard banded spatial splits, per marker chance levels), retiring an earlier leakage inflated result (PCC 0.88 to 0.96). Under it, IgG, C3, IgA, and ACTN4 recover above chance on unseen kidney cases, and one model per channel learns six antibodies across six mouse brain gels (**11 of 12 above chance**).
- Accelerated training **3x** with bf16 and torch.compile, and suppressed whole slide seam artifacts **3.7x** (3.70 to 1.01) through Hann weighted tile blending over a 38 slide ND2 to HDF5 pipeline.
- Designed a **segmentation to quantification** pipeline for four marker expansion microscopy of small cell lung cancer (DAPI, DLL3, SEZ6, CD276), deriving per cell membrane:cytoplasm ratios and N:C scores over **20+ compartment assignment algorithms** to show expansion resolves localization invisible before expansion.

Vyuhaa Med Data — Hyderabad, India
AI Integration Engineer — Oct 2024 – Jul 2025

- Shipped a **whole slide scanner** with on device AI, now **deployed in affiliated pathology labs**, after taking over **CerviAI** from a senior engineer's prototype and leading its three stage detection, classification, and segmentation engine onto an NVIDIA **Jetson Orin Nano**.
- Cut edge inference latency from **500s to 80s** by parallelizing inference and eliminating redundant OpenSlide handle initializations.
- Reached **90% precision** and a **10x gain in recall** on clinical grade WSI by migrating detection from YOLOv7 to YOLOv11.
- Reduced each retraining round to one command by closing an **active learning** loop with an in house pathologist, scripting false positives from review sheet to versioned dataset.

PROJECTS

Inv-SHAF: Donor Invariant Active Learning for Spatial Gene Expression from Histology — Apr 2026

- Predicted 100 gene **Visium** expression from H&E (30,868 spots, 8 donors) on **UNI2-h** embeddings, with a **domain adversarial** donor discriminator stripping batch effects.
- Beat uncertainty, core set, and random sampling at every label budget from 1% to 10% by formulating an **invariance violation acquisition function** that scores disagreement between an invariant learner and per donor cheater heads.

spotNUMT: Mitochondrial Pseudogene Sequence Classifier — Mar 2026

- Achieved **100% minority class recall** under a 1:33 class imbalance by training a CNN and BiLSTM ensemble to separate mitochondrial DNA from nuclear pseudogenes (NuMTs), deployed as a live Hugging Face app.

MitoGraph: Mitochondrial Knowledge Graph for VUS Pathogenicity Prediction — Feb 2026

- Scored 1,228 mitochondrial VUS across 808 phenotypes at **0.830 test AUPRC** with a heterogeneous knowledge graph and **GATv2Conv** attention network under a leakage free split.

MUFFLE: Multimodal Federated Learning on WSI and RNA-seq — Jan 2026

- Led an 11 person **NVFlare** team's multimodal fusion design, integrating WSI and RNA-seq encoders through gated attention to stratify patients into risk groups. **Best Collaboration Award**.

PUBLICATION

First author. Hybrid Sequential Architecture of CNN and Transformer for Diabetic Retinopathy Grading. Com-IT-Con 2024 (Taylor & Francis).